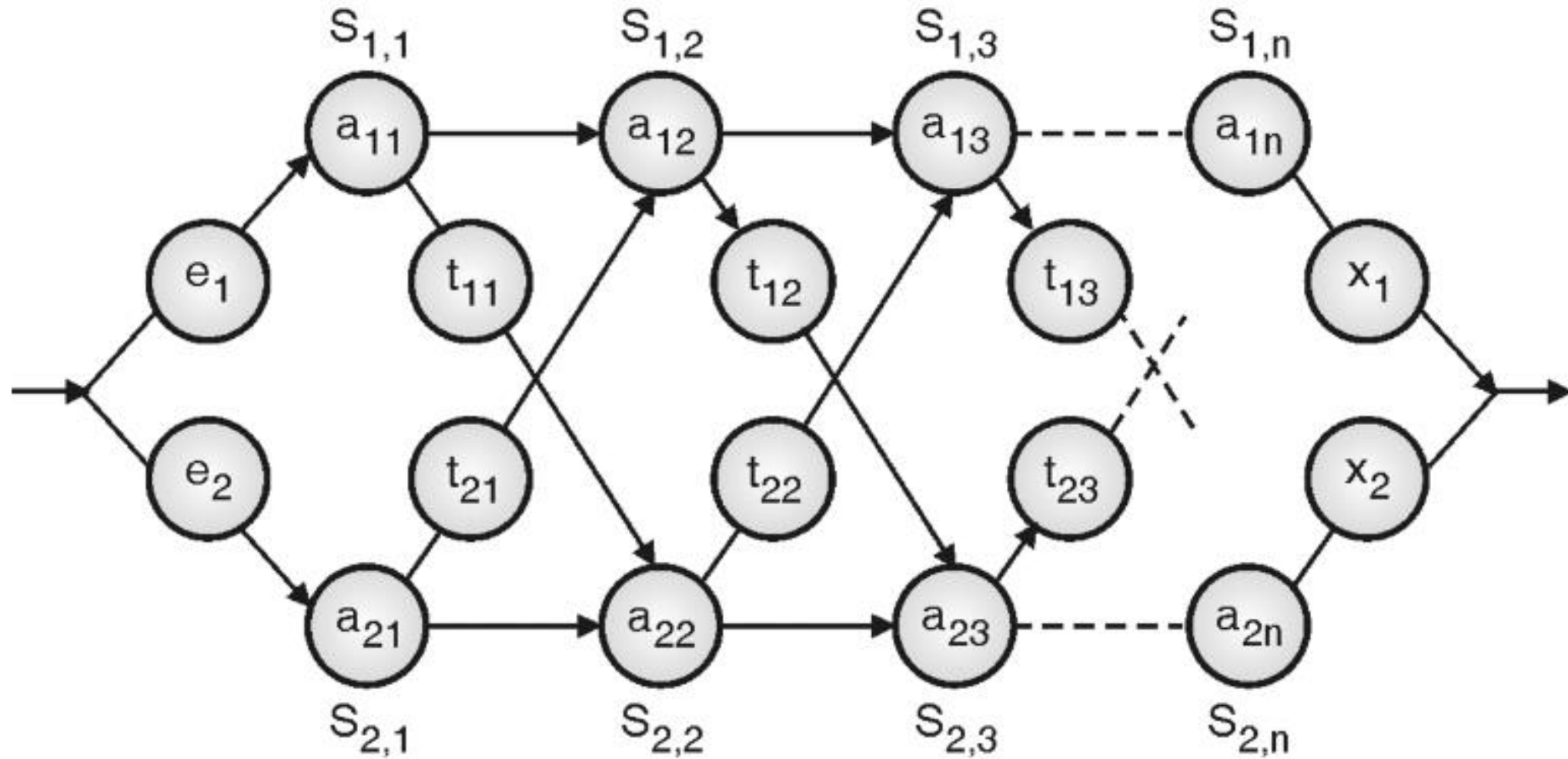


BCSE204L Design and Analysis of Algorithms

Module-2

Topic-4: Assembly Line Scheduling

Assembly Line Scheduling



Assembly Line Scheduling

- A factory has two assembly lines
- Each line has n stations $S_{1,1}$ to $S_{1,n}$ and $S_{2,1}$ to $S_{2,n}$ which builds partial products
- $S_{1,j}$ and $S_{2,j}$ perform the same function but can take different times $a_{1,j}$ and $a_{2,j}$
- Entry times are e_1 and e_2
- Exit times are x_1 and x_2

Assembly Line Scheduling

- After going through a station a partial product can stay on the same line with no cost or
- Transfer to the other line: Cost of transferring after $S_{i,j}$ is $t_{i,j}$
- Can be transferred to the other line if the load on the current line is high

Assembly Line Scheduling Problem

- Determine which station should be selected from assembly line 1 and which to choose from assembly line 2 in order to minimize the total time to build the entire product.

Formula

- Time required to build component on first station of both line is summation of entry time for particular assembly line and assembly time on first station of particular line.
- Initial condition (Entry Time) (*if* $j == 1$)
$$f_i[1] = e_i + a_{i1}; i = 1, 2$$

Formula

- $j > 1$ (Intermediate Times)
- $f_1[j] = \min (f_1[j - 1] + a_{1,j}, f_2[j - 1] + t_{2,j-1} + a_{1,j})$
- $f_2[j] = \min (f_2[j - 1] + a_{2,j}, f_1[j - 1] + t_{1,j-1} + a_{2,j})$
- Exit time (Objective function)
- $f^* = \min (f_1[n] + x_1, f_2[n] + x_2)$

Demo

- On board

Pseudocode-1

```
Algorithm ASSEMBLY_LINE_SCHEDULING(n, e, a, t, x)

// n is number of stations on both assembly
// e is array of entry time on assembly
// a is array of assembly time on given station
// t is array of the time required to change assembly
line

//x is array of exit time from assembly

f1[1]  $\leftarrow$  e1 + a1,1
f2[1]  $\leftarrow$  e2 + a2,1
```

Pseudocode-2

```
for j ← 2 to n do
    if  $f1[j - 1] + a1, j \leq f2[j - 1] + t[j - 1] + t +$ 
     $a2, j$  then
         $f1[j] \leftarrow f1[j - 1] + a1, j$ 
         $L1[j] \leftarrow 1$ 
    else
         $f1[j] \leftarrow f2[j - 1] + t2, j - 1 + a1, j$ 
         $L1[j] \leftarrow 2$ 
    end
```

Pseudocode-3

```
    if  $f2[j - 1] + a2, j \leq f1[j - 1] + t1, j - 1 + a2,$   
 $j$  then  
         $f2[j] \leftarrow f2[j - 1] + a2, j$   
         $L2[j] \leftarrow 2$   
  
    else  
         $f2[j] \leftarrow f1[j - 1] + t1, j - 1 + a2, j$   
         $L2[j] \leftarrow 1$   
  
    end
```

Pseudocode-4

```
    if  $f1[n] + x1 \leq f2[n] + x2$  then  
         $F^* \leftarrow f1[n] + x1$   
         $L^* \leftarrow 1$   
    else  
         $F^* \leftarrow f2[n] + x2$   
         $L^* \leftarrow 2$   
    end  
end
```

Pseudocode- Print Stations

```
Algorithm PRINT_STATIONS(l, n)
```

```
    i ← 1*
```

```
    print "line " i ", station " n
```

```
    for j ← n downto 2 do
```

```
        i ← Li[ j ]
```

```
        print "line " i ",station " j - 1
```

```
    end
```

Analysis

- A single for loop
- So, $O(n)$

Thank you..